

Headaches

Principles and Practices of Osteopathic Medicine I

Adam Reno, DO, MPH
Assistant Professor, Clinical Medicine
Arkansas Campus
areno@nyit.edu

Session Objectives

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

- Define primary and secondary types of headaches
- Distinguish between primary and secondary headaches according to the international classification of headache disorders
- Identify different types of headaches based on their presentation, symptoms, and causes
- Identify “Red Flags” in the evaluation of headaches
- List treatment options based on the type of headache

<https://exchange.scholarrx.com/brick/m645g1q65jq1>

<https://exchange.scholarrx.com/brick/secondary-headaches>

Headache: A Disabling Problem

The Burden of Headache

Symptom Burdens

During Attacks

- Pain & suffering leading to reduced functional ability

Between Attacks

- Fear of next attack leading to avoidance behaviors and lifestyle compromise

Secondary Burdens

During Attacks

- Loss of work-time/work-effectiveness
- Failed or abdicated social roles
- Curtailed social and leisure activities

Beyond Attacks

- Failure to meet full potential
- Cessation of social planning
- Breakdown of relationships

Headache: Overview

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

- Headache is common & highly disabling
- Headache is often misdiagnosed and under-treated
- Effective treatment is often possible but hinges on correct diagnosis
- Headache disorders have a large differential diagnosis so need to be approached systematically
- [The Revised International Classification of Headache Disorders \(ICHD-3\)](#) by the International Headache Society (IHS) provide a framework for headache diagnosis

[Scholar Rx Brick: Migraine, Cluster, and Tension Headaches](#)

Headache: International Headache Society

Primary Headaches

- Tension-Type Headache
- Migraine
- Cluster Headache and Other Trigeminal Autonomic Cephalgias

Secondary Headaches

- Infection
- Head Injury
- Vascular Disorders/Bleeds
- Brain Masses/Tumors

<https://exchange.scholarrx.com/brick/m645g1q65jq1>

Primary Headaches

- Benign and recurrent
- NOT usually associated with underlying pathology
- The headache is the disease

[Scholar Rx Brick: Migraine, Cluster, and Tension Headaches](#)

Primary Headaches

Migraine

- With Aura
- Without Aura

Tension-Type Headache

- Episodic
- Chronic

Cluster Headache and Other Trigeminal Autonomic Cephalalgias

- Cluster headache
- Paroxysmal hemicrania
- Short unilateral neuralgiform headache with conjunctival injection and tearing (SUNCT)
- Short-lasting unilateral neuralgiform headache attacks with autonomic symptoms (SUNA)
- Hemicrania continua



Secondary Headaches

- Can be sudden or progressive
- Usually associated with pathology
- May require immediate action

[Scholar Rx Brick: Migraine, Cluster, and Tension Headaches](#)

Secondary Headaches:

- IHS Classification

Headache attributed to:

- Head and/or neck trauma
- Cranial or cervical vascular disorders
- Nonvascular intracranial disorders
- Substance or withdrawal thereof
- Infection
- Disorder of homeostasis
- Disorder of cranium, neck, eyes, ears, nose, sinuses, teeth, other facial structures
- Psychiatric

Secondary Headaches

Examples

- Aneurysms
- AVMs and subarachnoid hemorrhage
- Meningitis
- Cerebrovascular Accident
- Intracranial Neoplasia
- Seizure
- Chiari Malformation
- Glaucoma
- Trauma
- Temporal Arteritis
- Hypertension
- Lumbar Puncture Headache
- Sinuses
- TMJ Disorder
- Medication Overuse Syndrome
- Toxins/Poisons
- Idiopathic Intracranial Hypertension

[Scholar Rx Brick: Migraine, Cluster, and Tension Headaches](#)

Cranial Neuralgias and other headaches: IHS Classification

Cranial neuralgias, central and primary facial pain

- Trigeminal neuralgia
- Glossopharyngeal neuralgia
- Other terminal branch neuralgias
- Cold stimulus headache
- External compression headache

Other headaches

- Headache, not elsewhere classified
- Headache, unspecified

Headache Assessment

Purpose of assessment

- Exclude any urgent, emergent, and/or serious cause
- Make an accurate diagnosis of headache type
- Identify exacerbating/aggravating factors
- Optimize treatment (acute and preventative)
- Explain diagnosis and rationale for treatment

Assessment Tools

- History taking
- Physical exam
- Headache diary
- Imaging studies
- Laboratory analysis

Headache Assessment

History Taking

NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine

A thorough history is ***the*** single most useful tool for defining headache diagnosis and initiating a treatment plan.

Headache Assessment

History Taking - Key Questions

Time

- Onset, frequency, patterns, duration

Character

- Description, intensity, nature of pain

Location

- Localized, diffuse, radiating, laterality

Causes

- Predisposing, triggering, exacerbating, receiving factors

Response

- Patient's actions and limitations during headache
- Medications used and how often

Associated symptoms

- Auras, nausea, vomiting, visual changes, photo/phonophobia, lacrimation, jaw pain

Headache History

- Does the patient take any medications?
 - Prescription, alternative/herbal, over-the-counter
- Does the patient have any other medical conditions?
 - HIV, cancer, hypertension, diabetes, hypothyroidism, seizures, etc.
- Surgical history
- Allergies
- Social history
 - Smoking, EtOH use, drug use, caffeine use
 - Stress/anxiety level
 - Sleeping habits
 - Eating habits
 - Exercise habits

Headache Triggers

- Diet
 - EtOH
 - Chocolate
 - Aged cheese
 - MSG
 - Aspartame
 - Caffeine
 - Nuts
 - Low H2O intake
- Hormones
 - Menses
 - Ovulation
 - Oral contraceptives
 - Hormone replacement therapy
- Stress/Anxiety
 - Loss or change:
 - Death
 - Divorce
 - Moving/relocation
 - Crisis period
 - Intense activity
 - Let-down period
- Change in habits
 - Weather
 - Travel
 - Diet/sleep
 - Physical activity

Physical Exam for Headache

- Blood pressure and other vital signs including temperature
- BMI
- Complete HEENT exam
 - Temporal artery inspection and palpation
 - Signs of upper respiratory infection
- Neck exam
 - Meningeal signs
 - Auscultation for carotid bruit
- Standard physical exam
- Neurologic exam: mental status, cranial nerves including visual acuity and fundoscopy, motor, sensory, coordination and gait
- Depression screening

Headache diary

Date:	Date:	Date:
Warning signs:	Warning signs:	Warning signs:
Time begun:	Time begun:	Time begun:
Time ended:	Time ended:	Time ended:
Type of pain: eg, piercing, throbbing, etc.	Type of pain: eg, piercing, throbbing, etc.	Type of pain: eg, piercing, throbbing, etc.
Intensity of pain: Circle one Low 1 2 3 4 5 6 7 8 9 High	Intensity of pain: Circle one Low 1 2 3 4 5 6 7 8 9 High	Intensity of pain: Circle one Low 1 2 3 4 5 6 7 8 9 High
Location: eg, between eyes, back of head, etc.	Location: eg, between eyes, back of head, etc.	Location: eg, between eyes, back of head, etc.
Treatment or medication taken:	Treatment or medication taken:	Treatment or medication taken:
Effect of treatment:	Effect of treatment:	Effect of treatment:
Hours of sleep:	Hours of sleep:	Hours of sleep:
What I ate today:	What I ate today:	What I ate today:
Events prior to headache: eg, strenuous activity, elevated stress, etc.	Events prior to headache: eg, strenuous activity, elevated stress, etc.	Events prior to headache: eg, strenuous activity, elevated stress, etc.
Comments:	Comments:	Comments:

Reproduced with permission from: Bristol-Myers Squibb Company. Copyright 2001 Bristol-Myers Squibb Company.

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic
Medicine

Migraine Headache

- Affects 12-15% of the general population
- Female > males
 - Up to 17% in females
 - Up to 6% in males
- Family history (present in 70% of migraine cases)
- Aura symptoms around 5%

Migraine Headache

- Pathophysiology -

- Primarily a neuronal dysfunction
- Initially thought to be due to dilatation of blood vessels, however proven to be incorrect
- Cortical spreading depression (Cortical Spreading Depression of Leao)
 - Self-propagating wave of neuronal and glial depolarization spreading across cerebral cortex
 - Cortical spreading depression leads to:
 - Cause the aura of migraine
 - Activate trigeminal nerve afferents
 - Alter blood-brain barrier permeability

Migraine Headache

- Phases and Symptoms -

- Four phases of migraine “attack”:
 - Prodrome
 - Aura
 - Pain
 - Postdrome

Migraine Headache: Prodrome

<u>Psychological</u>	<u>Neurological</u>	<u>Constitutional and autonomic</u>
Depression	Photophobia	Stiff neck
Hyperactivity	Difficulty concentrating	Food craving
Euphoria	Phonophobia	Cold feeling
Talkativeness	Dysphasia	Anorexia
Irritability	Hyperosmia	Sluggishness
Drowsiness	Yawning	Diarrhea or constipation
Restlessness		Thirst
		Urination
		Fluid retention

Migraine Headache: Aura

- Appears over 5 to 20 minutes and usually lasts less than 60 minutes
- Aura symptoms can be visual, sensory, or motor in nature
- Visual aura is the most common type of neurological event (90%)

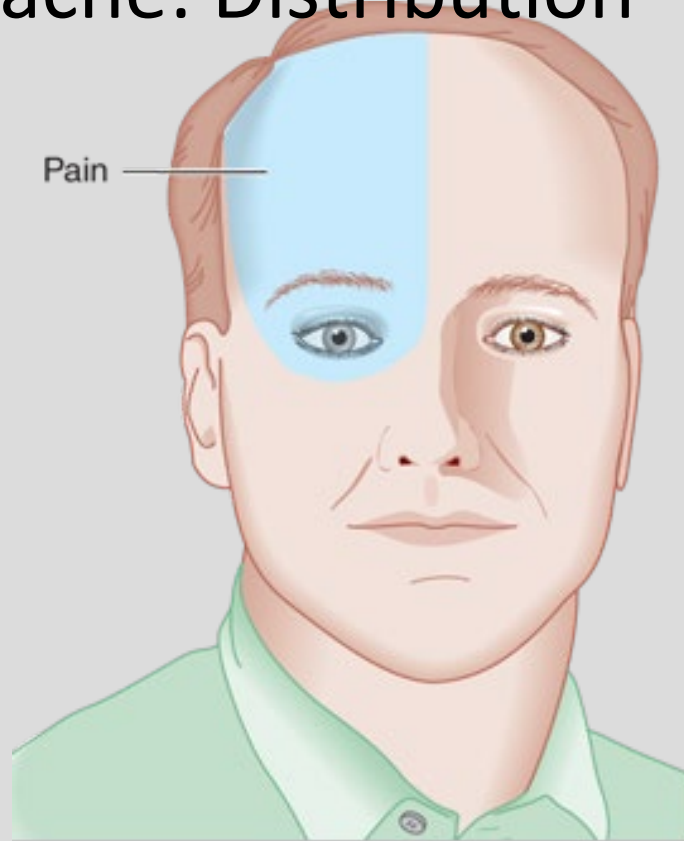
Migraine Headache: Headache/Pain

NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine

- Typically unilateral, throbbing, moderate to severe pain, and aggravated by physical activity
- Usually gradual onset
- Pain usually peaks and then subsides:
 - 4 to 72 hours in adults
 - 1 to 48 hours in children
- Usually accompanied by nausea (90%) and vomiting (30%)
- Photophobia, Phonophobia, Osmophobia

Migraine Headache: Distribution



**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

<https://exchange.scholarrx.com/brick/m645g1q65jq1>

Migraine Headache: Postdrome

- Tired, listless, irritable, mood changes, scalp tenderness
- Some patients will feel unusually refreshed or euphoric, whereas others report depression or malaise
- Often, some symptoms of the headache phase can continue

Migraine Headache: Without Aura

Diagnostic Criteria

Attacks lasting 4-72 hours, having occurred at least 5 times

At least *two* of the following:

- Unilateral
- Pulsating
- Moderate to severe
- Aggravated by exercise

At least *one* of the following:

- Nausea
- Photophobia
- Phonophobia

Migraine Headache: With Aura

Diagnostic Criteria

- Migraine without aura PLUS
 - One or more transient focal neurological signs:
 - Visual
 - Sensory
 - Aphasia
 - Motor
- Gradual development of aura symptoms
- Aura lasts 4 to 60 minutes followed by or accompanied by headache within 60 minutes

Migraine Headache

- Predisposing Factors/Triggers -

- Family history
- H/o travel sickness
- H/o cyclical vomiting in childhood
- Stress
- Menstruation
- Hormonal replacement therapy
- Oral contraceptive pills
- Head/neck trauma
- Strenuous exercise
- Relaxation after stressful events
- Change of habits
 - Missed meals, missed sleep
- Bright lights
- Loud noises
- Smells
 - Perfumes, gasoline
- Dietary (<10%)
 - Chocolate
 - Cheese
 - Aspartame
 - EtOH

Migraine Headache

- Acute Treatment -

Basic Principles

- Correct dose
- Correct formulation
- Step-up treatment
 - Simple analgesia —> combinations —> triptans/ergots
- Early intervention
- Use at least 3 times before judging effectiveness
- Use a maximum of 2-3 days per week or 8 days in a month

Migraine Headache

- Acute Treatment -

- Analgesics
 - Acetaminophen
 - NSAIDs
- Specific anti-migraine therapies
 - Triptans (5HT- agonist (sumatriptan, Rizatriptan, etc.)
 - Ergots (Ergotamine)
- Anti-emetics
 - Promethazine (Phenergan)
 - Prochlorperazine (Compazine)
 - Metoclopramide
- Combinations
 - Excedrin (Acetaminophen/aspirin/caffeine)



**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine



Migraine Headache

- Management and Prevention -

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

- Lifestyle modifications
 - Regular sleep
 - Regular meals
 - Moderate amount of exercise
 - Drink plenty of water
 - Limit caffeine, alcohol, and other drugs
 - Reduce stress
- Cognitive behavioral therapy
- Biofeedback
- Relaxation training
- OMT
- Acupuncture
- Pharmacological treatments

Migraine Headache

- Management and Prevention -

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

When to Use

- Frequent headaches - >3-4/month
- Migraine significantly interferes with patients daily life despite abortive treatment
- Acute/abortive therapies contraindicated, ineffective, not tolerated, or OVERUSED

Migraine Headache

- Prescription Treatments -

- Antihypertensives
 - Beta-blockers (propranolol, nadolol)
 - Calcium-channel blockers (Verapamil, amlodipine)
- Antidepressants
 - TCAs (amitriptyline)
 - 5HT/NE re-uptake inhibitors (venlafaxine)
- Anticonvulsants
 - Valproate
- CGRPs (Nurtec, Amovig, Emgality, Ubrovelvy)

Migraine Headache

- General Principles -

- Consider comorbidities and lifestyle factors
- Titrate dose - “start low and go slow”
- Aim for optimal dose
- Try medication for at least 3 months
- Some patients may relapse and require long-term therapy



NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic
Medicine

Tension-Type Headache

- **Most common type of headache!**
- Women > men
- Mild to moderate in severity
- Genetics may play a minor role
- Lasting as little as 30 minutes to as long as 7 days
- Myofascial tenderness is increased
- Often coexists with migraine headache
- Triggers - stress, fatigue, lack of sleep
- Comorbid conditions - depression and anxiety in >50% of patients

Tension-Type Headache

- Pathophysiology -

- Precise mechanism is not certain, likely multifactorial
- Peripheral activation or sensitization of myofascial nociceptors - episodic
- Sensitization of pain pathways in the central nervous system due to prolonged nociceptive stimuli from pericranial myofascial tissues - chronic
- Nitric oxide may play a role in tension-type headaches - currently under study
- Genetic factors may play a role for chronic, not episodic tension-type headache

Tension-Type Headache

- IHS Criteria -

- Require at least 10 episodes of headache, each lasting 30 minutes to seven days, which fulfill the following conditions:

At least two of the following:

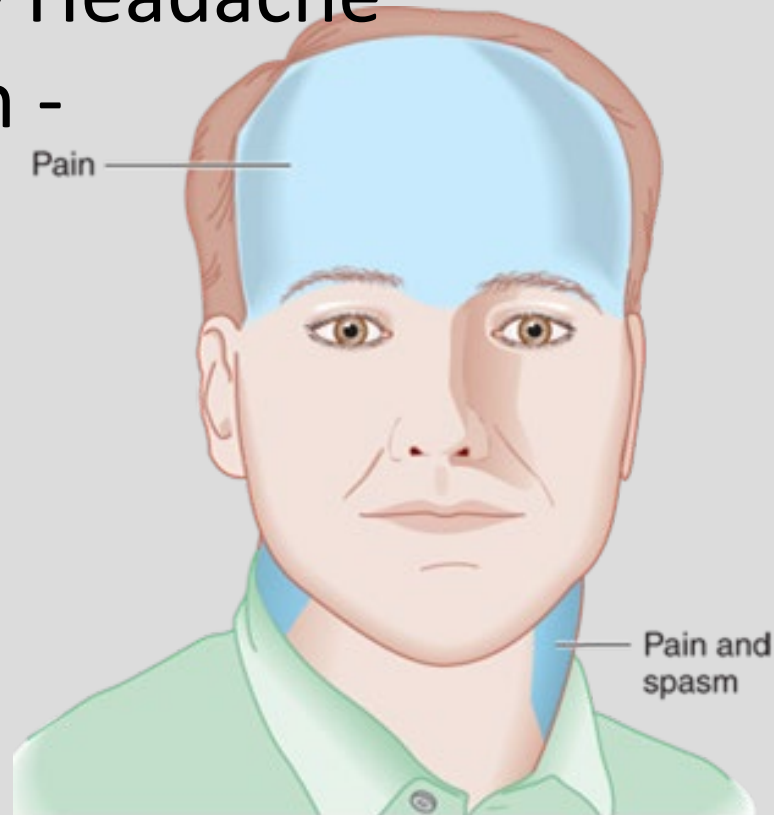
- Bilateral location
- Pressing or tightening (non-pulsating) quality
- Mild or moderate intensity
- Not aggravated by routine physical activity such as walking or climbing stairs

Both of the following:

- No nausea or vomiting
- No more than one of photophobia or phonophobia

Tension-Type Headache

- Distribution -



**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

<https://exchange.scholarrx.com/brick/m645g1q65jq1>

Tension-Type Headache

Episodic Tension-Type Headache

- Episodic TTH (*<15 headaches a month*) - initial treatment with simple analgesics
 - NSAIDs or Aspirin
 - Acetaminophen
 - Parenteral Medications
 - Ketorolac, Metoclopramide, Chlorpromazine
 - Limited Evidence - used in severe cases

Chronic Tension-Type Headache

- Chronic TTH (*≥15 headache days a month*), *for at least 6 months*
 - Acute symptoms - NSAIDs or acetaminophen
 - Amitriptyline, skeletal muscle relaxants, fluoxetine
 - Stress managements, OMT, physical therapy, massage, acupuncture

Cluster Headache

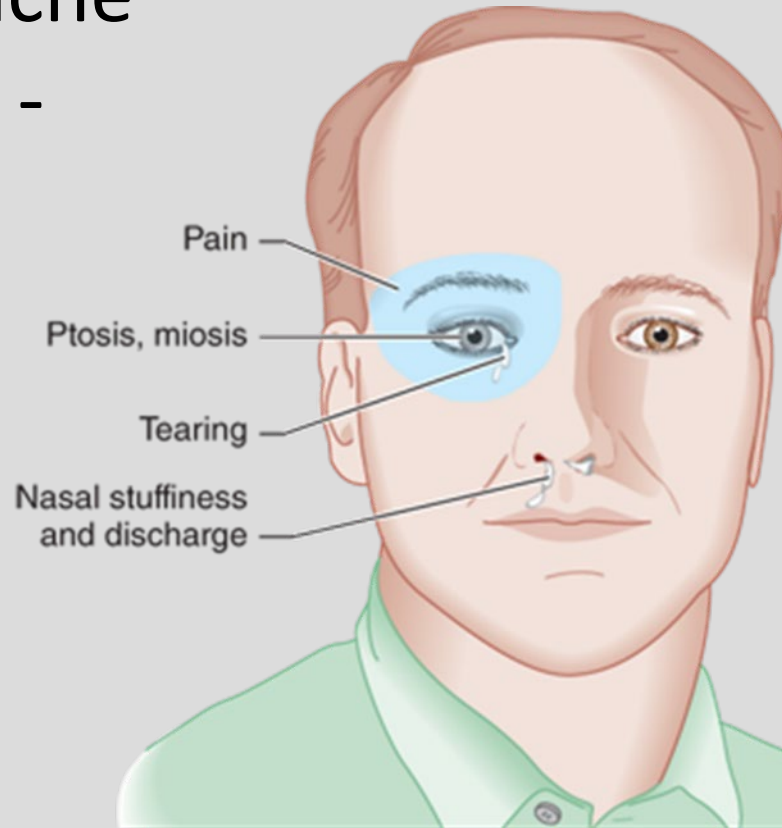
- Age of onset 25-50 years, Male > Female
- Diagnostic criteria:
 - Attacks “clustered” in time (>5)
 - Severe unilateral, orbital supraorbital, and/or temporal pain
 - Duration of 15 to 180 minutes, occurring 1 to 8 times per day
 - Associated with one of the following ipsilateral signs:
 - Conjunctival injection/lacrimation
 - Nasal congestion/rhinorrhea
 - Eyelid edema
 - Forehead/facial swelling
 - Miosis/ptosis (possibly both)
 - Restlessness/agitation

Cluster Headache

- Distribution -

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine



<https://exchange.scholarrx.com/brick/m645g1q65jq1>

Cluster Headache

- Pathophysiology -

- Complex and remains incompletely understood
- Most widely accepted theory is hypothalamic activation with secondary activation of the trigeminal-autonomic reflex, likely via the trigeminal-hypothalamic pathway
- Another accepted theory is neurogenic inflammation of the walls of the cavernous sinus obliterates venous outflow, which injures the traversing sympathetic fibers of the intracranial internal carotid artery and its branches

Cluster Headache

- Treatment
 - Acute:
 - 100% Oxygen, subcutaneous or nasal triptans
 - Chronic/Preventative:
 - Verapamil, lithium, corticosteroids, anticonvulsants

Characteristics of common headache syndromes

Symptom	Migraine headache	Tension headache	Cluster headache
Location	Unilateral in 60 to 70 percent; bifrontal or global in 30 percent	Bilateral	Always unilateral, usually begins around the eye or temple
Characteristics	Gradual in onset, crescendo pattern; pulsating; moderate or severe intensity; aggravated by routine physical activity	Pressure or tightness which waxes and wanes	Pain begins quickly, reaches a crescendo within minutes; pain is deep, continuous, excruciating, and explosive in quality
Patient appearance	Patient prefers to rest in a dark, quiet room	Patient may remain active or may need to rest	Patient remains active
Duration	4 to 72 hours	Variable	30 minutes to 3 hours
Associated symptoms	Nausea, vomiting, photophobia, phonophobia; may have aura (usually visual, but can involve other senses or cause speech or motor deficits)	None	Ipsilateral lacrimation and redness of the eye; stuffy nose; rhinorrhea; pallor; sweating; Horner's syndrome; focal neurologic symptoms rare; sensitivity to alcohol

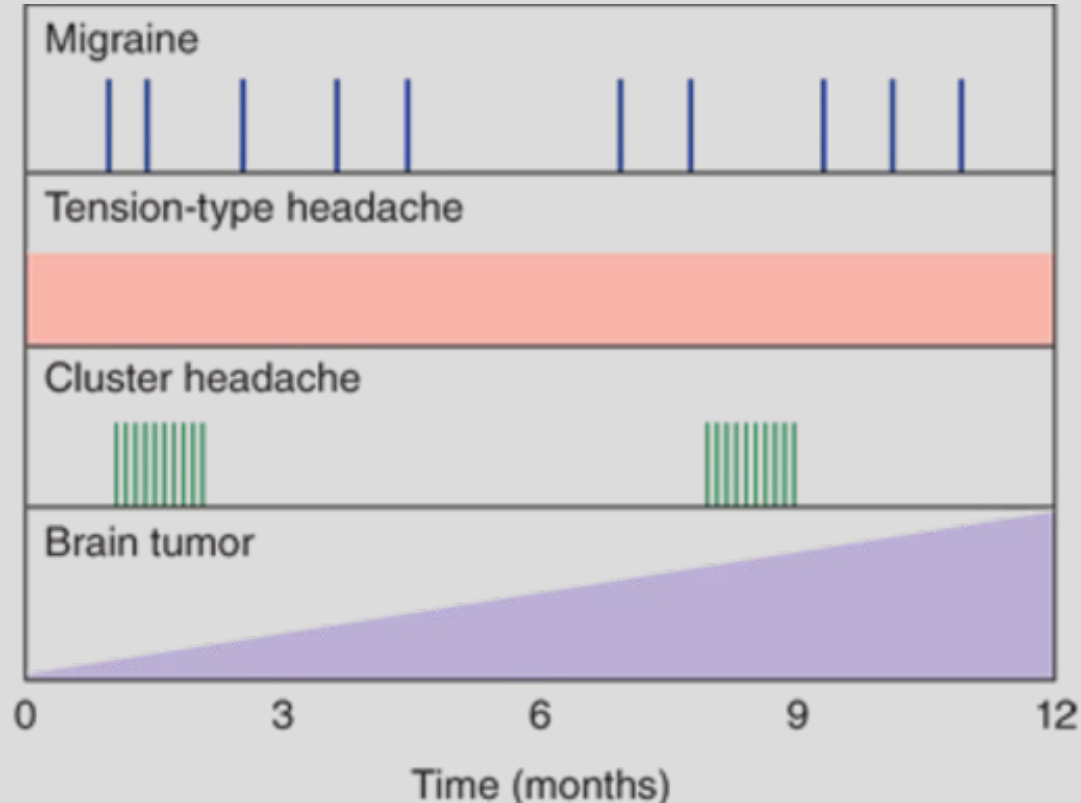
NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic
Medicine

Headache Patterns

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine



Secondary Headache Examples

- Aneurysms, AVMs, subarachnoid hemorrhages
- Meningitis
- Cerebrovascular accidents
- Intracranial neoplasia
- Seizure
- Chiari malformation I
- Glaucoma
- Trauma
- Temporal arteritis
- Dialysis headache
- Hypertension
- Lumbar puncture headache
- Sinus headache
- Temporal mandibular arthralgia
- Medication overuse syndrome
- Toxic/poison ingestion
- Idiopathic intracranial hypertension

Secondary Headache - RED FLAGS -

- Sudden onset, worst headache of my life (thunder-clap)
- Onset after age 50
- History of recent trauma
- Currently anti-coagulated
- Systemic signs (fever, myalgias, weight loss)
- Systemic disease (malignancy, HIV)
- Headache with change in posture
- Headache with straining (Valsalva, coughing)
- Change in frequency and severity
- Neurological symptoms or abnormal physical exam findings
- Altered mentation and focal neurological findings

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

Secondary Headaches

- Subarachnoid Hemorrhage
- Intracerebral Hemorrhage
- Meningitis or Encephalitis
- Lumbar puncture
- Giant Cell (Temporal) Arteritis
- Intracranial Mass (tumor, subdural hematoma, abscess)

Secondary Headache

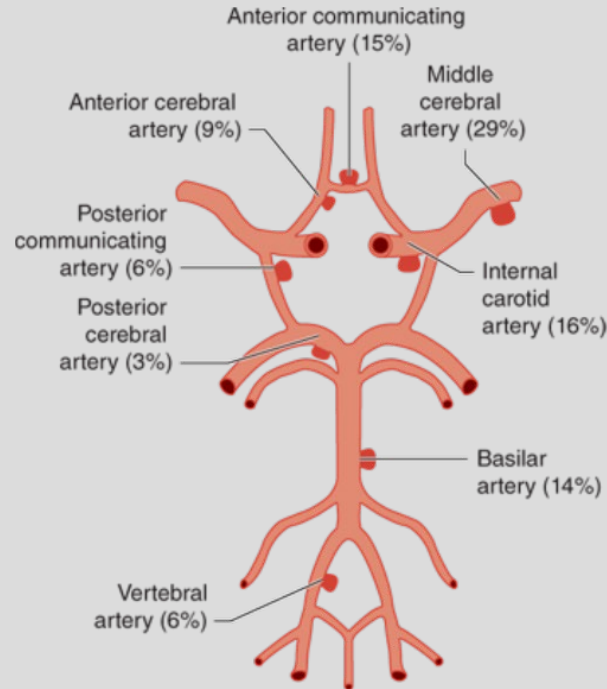
- Subarachnoid Hemorrhage -

- Usually, the result of a ruptured cerebral arterial aneurysm or an AVM
- ~75% of time it is a saccular (berry) aneurysm
- Rupture occurs most often during ages 50's to 60's, with an equal sex distribution
- Most arise sporadically, but some are familial
- Classic: sudden onset, unusually severe generalized headache, described as “the worst headache of my life” (thunderclap)
- Computed tomography (CT) scan
- BP control / surgery

Secondary Headaches: Intracranial Aneurysms- Frequency Distribution

**NEW YORK INSTITUTE
OF TECHNOLOGY**

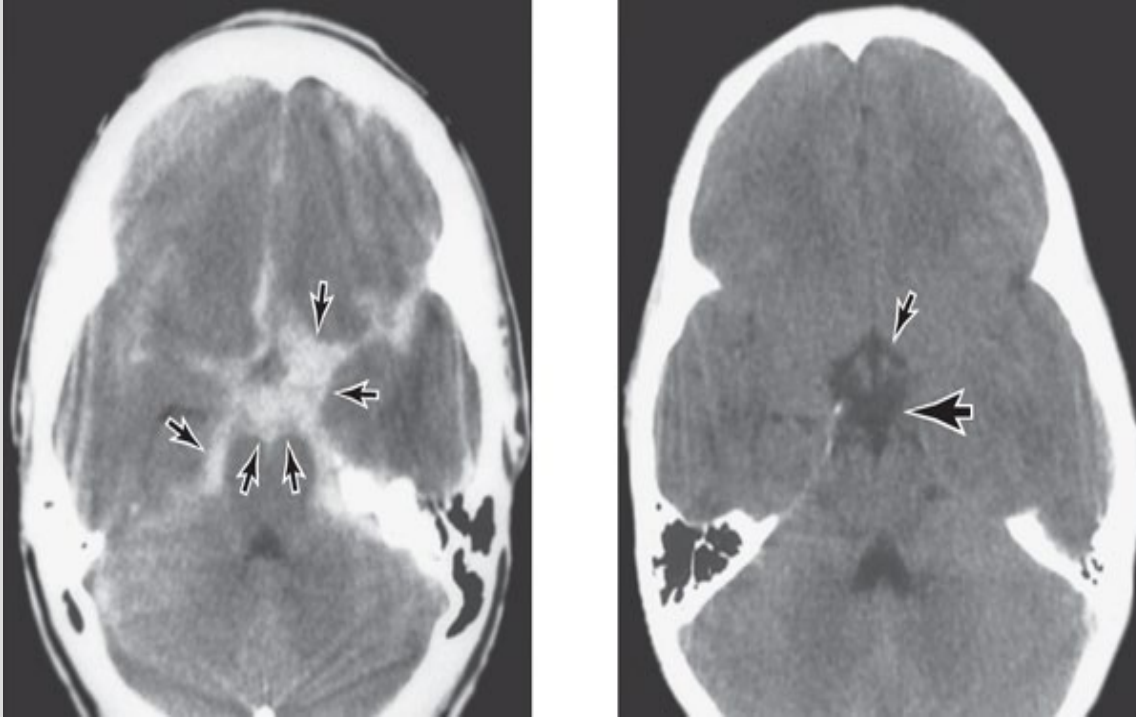
College of Osteopathic
Medicine



Secondary Headaches: Intracranial Aneurysms- CT Scan of Head

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine



Secondary Headaches

- Intracerebral Hemorrhage -

- Headache, vomiting, altered consciousness, and focal neurologic deficits.
- Compression of pain-sensitive structures by the hematoma.

Secondary Headaches

- Meningitis or Encephalitis -

- Combination of new headache (87%), neck stiffness (83%), fever (77%), and altered mental state (69%)
- Infection involving brain parenchyma (encephalitis) presents similarly with headache (81%), fever (90%), and altered consciousness (97%), but seizures are frequent (67% in encephalitis while only 5% in meningitis).
- Pain is caused by inflammation of intracranial pain-sensitive structures including blood vessels at the base of the brain.
- Commonly throbbing in character, bilateral, and occipital or nuchal in location.

Secondary Headaches

- Lumbar Puncture -

- History of a dural puncture (e.g. spinal tap, spinal anesthesia, etc.) and is characteristically a postural headache with increase in pain in the upright position and relief with recumbence.
- Presentation:
 - ✓ occipital
 - ✓ within 48 to 72 hours after the procedure
 - ✓ lasts 1 to 2 days.
- Treatment: Blood patch, Caffeine

Secondary Headaches

- Giant Cell (Temporal) Arteritis -

- Systemic vasculitis that affects medium-sized and large arteries (temporal), especially branches of the external carotid artery
- Inflammation of the pain-sensitive arterial wall produces headache, and arterial stenosis leads to ischemia
- Women > men and is uncommon before 60 years of age
- Most common symptom is a new, nonspecific headache, unilateral or bilateral and is often fairly severe and boring in quality
- ESR is elevated, but not invariably; requires prompt evaluation to avoid vision loss
- Diagnosis made by biopsy of an affected temporal artery, which is characteristically thickened, nonpulsatile, dilated, and tender

Secondary Headaches

- Intracranial Mass-

- New onset of headache in middle or late life should always raise concern
- 50% of patients with brain tumor complain of headache
- Headaches associated with brain tumors are most often nonspecific in character, mild to moderate in severity, dull and steady in nature, and intermittent.
- Characteristically bifrontal, worse ipsilaterally, and aggravated by a change in position or by maneuvers that increase intracranial pressure, such as coughing, sneezing, and straining at stool.
- Classically maximal on awakening in the morning and is associated with nausea and vomiting.

In Summary...

- Define primary and secondary types of headaches
- Distinguish between primary and secondary headaches according to the International Classification of Headache Disorders and International Headache Society
- Identify the different types of headaches based on their presentation/symptoms/causes
- Identify “RED FLAGS” in the evaluation of headaches
- List treatment options based on type of headache

Resources and References

[Scholar Rx Brick: Migraine, Cluster, and Tension Headaches](#)

[Scholar Rx Brick: Secondary Headaches](#)

Additional Required Resources:

[Headache & Facial Pain](#). In: Simon RP, Aminoff MJ, Greenberg DA. eds. *Clinical Neurology, 10e* New York, NY: McGraw-Hill.

Suggested/Supplemental Resources:

[The Revised International Classification of Headache Disorders \(ICHD-3\)](#)

[Goadsby, Peter J.. "Headache." Harrison's Principles of Internal Medicine, 20e](#) Eds. J. Larry Jameson, et al. New York, NY: McGraw-Hill.

Lecture and Lab Feedback Form:

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>